

1.4 Achievements

During the development of the Smart Event Management and Ticket Booking System, the following achievements have been made:

- Completely developed a centralized web-based event management and ticket booking system.
- Implemented secure User Registration/Login.
- Implemented administrator authentication and role-based access.
- Developed event management functionalities for administrators.
- Implemented a system that allows users to reserve tickets online.
- Implemented a simulated payment management system.
- Stored data on the cloud using MongoDB Atlas.
- Implemented JWT-based authentication and bcrypt password encryption.
- Developed REST APIs using Node.js and Express.js.
- Implemented protected routes and authorization systems.
- Carried out unit, integration, and system testing.
- Successfully deployed the frontend using Vercel.
- Successfully deployed the backend using Railway.
- Connected the frontend, backend, and MongoDB Atlas database.
- Designed an application that is user-friendly and intuitive to use.

1.5 Organization of Report

The project report consists of various chapters, with each chapter explaining a different stage in the development of the Smart Event Management and Ticket Booking System.

Chapter 1 – Introduction

This chapter reflects the background of the project, statement of the problem, objectives, scope, project achievements, and organization of the project report.

Chapter 2 – Survey of Technologies

This chapter explains the technologies, frameworks, tools, and platforms utilized to build the system. It includes React.js, Vite, Node.js, Express.js, MongoDB Atlas, JWT, bcrypt, Git, GitHub, Vercel, and Railway.

Chapter 3 – Requirements and Analysis

This chapter explains the requirements and analysis of the system. It covers the functional and non-functional requirements, hardware and software requirements, project planning, and analysis of the existing and proposed systems.

Chapter 4 – System Design

This chapter presents the design of the proposed system. It comprises system design, database design, data integrity and constraints, security considerations, UML diagrams, and other design-related components.

Chapter 5 – Implementation and Testing

This chapter explains the approach taken to implement the system. It includes input and output design, database implementation, table structures, code modules, system implementation, and testing approaches such as unit testing, integration testing, and system testing.

Chapter 6 – Modules

This chapter covers the objectives, inputs, processing, outputs, validation rules, and expected outcomes of the various modules implemented in the system. It explains the major modules, including Event Management, Booking Management, Payment Management, and User Management and Authentication.

Chapter 7 – Conclusion and Future Scope

This chapter provides the conclusion of the project, limitations of the system, and future work that can be carried out to enhance the system and make it more suitable for future applications.